

Canonical admission and decoder repair

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Result at a glance

[DERIVED] One repair mechanism is sufficient at the frozen interface: canonical exact encodings, prime-power exact-order tests, a full-order Weil pairing test for the ordered $E[2^a]$ basis, and a strict separation between public pre-admission and post-evaluation decoder admission.

[DERIVED] On every admitted honest relation, exact ciphertext equality is injective in (m, r) , and a semantically correct pair $A_0=f(U)$, $B_0=f(V)$ decodes the actual selected message. **[PROPOSED / OPEN]** The exact calibrated recipient map and a uniform two-point evaluator remain separate obligations. An autonomous base generator for the inherited complete evaluable $\text{End}(E)$ representation also remains open.

[MEASURED] There are no measured data, samples, uncertainty intervals, or run IDs. Scientific, implementation, formalizer, and experiment run counts are all zero. **[INDEPENDENTLY VERIFIED]** Nothing in this producer round; the frozen review plan assigns that work to later independent tasks.

Question and exact identities

The question was whether the $P=0$ and $V=0$ defects can be repaired without smuggling in the missing evaluator. The exact symbolic-family slice is:

- $p>3$, $p \equiv 3 \pmod{4}$, $F_{(p^2)}=F_p[i]/(i^2+1)$;
- $D=2^a$, $N=5^b$, $a \geq 2$, $b \geq 1$, $N > 4D$, with exact corresponding valuations in $p+1$;
- the exact source model $E: y^2=x^3+x$ and an exact canonical codomain C ;
- exact-order P in $E[N]$, X in $C[N]$;
- a role-tagged ordered basis (U, V) of $E[D]$;
- an honest normalized separable degree- D map $f: E \rightarrow C$ with $\ker(f) = \langle U+sV \rangle$, $s=m+2^k r$, and full signed $f(P)=X$.

There are no concrete sampled endpoints. Curve equality means complete field/model/coefficient bytes; point equality includes full signed y .

Methods

[DERIVED] The work was a zero-run symbolic derivation. It reproduced the prior $P=0$ and $V=0$ collision fibers, replaced membership by exact prime-power orders, used the Weil-pairing determinant criterion for a basis, typed the two admission stages, derived bounded honest rejection sampling, proved exact relation injectivity, and carried correct evaluator outputs through a checked power-of-two digit decoder.

Internal source links are collected under “Sources and provenance.” No external primary source was fetched. The KB items were used only at their stated reported/abstract tier, not as proof of this derivation or of novelty.

The staged predicate

[DERIVED] For $N=5^b$, exact order is $[N]T=0$ and $[N/5]T!=0$. For $D=2^a$, it is $[D]T=0$ and $[D/2]T!=0$. Apply these to P, X and U, V , respectively.

[DERIVED] Compute $w=e_D(U, V)$ and require exact order D . In coordinates relative to any $E[D]$ basis, w is a primitive root raised to the coordinate determinant; exact order means the determinant is odd, hence invertible modulo 2^a . This makes (U, V) an ordered basis and excludes $V=0$ and dependent pairs.

PreAdmit checks only canonical parameters/models, curve and point membership, exact P, X, U, V orders, and the pairing basis. It returns a receipt binding all full bytes. PostAdmitDecode is unavailable until a separate evaluator returns $A0, B0$; it checks their exact target model, $[D]A0=0$, $\text{ord}(B0)=D$, and $\text{ord}([2^{(a-k)}]B0)=2^k$ before decoding.

The complete predicate is admission-predicate.yaml, and exact serialization is exact-model-encodings.yaml.

Honest compatibility and retry accounting

[DERIVED, conditional on BaseGen] On the exact source curve, $E(F_{(p^2)})=E[p+1]$. Exact cofactor projections give uniform $E[N]$ and $E[D]$ samples. An $E[N]$ point is a generator with probability $24/25$; an ordered pair in $E[D]^2$ is a basis with probability $3/8$. With the bounded exact uniform-point sampler, the KeyGen failure bound is

$$\text{epsilon_base} + (t_P + 2t_B)2^{-t_R} + (1/25)^{t_P} + (5/8)^{t_B}.$$

For encryption $K=U+sV$ has exact order D because its U coefficient is one. An a -step normalized degree-two quotient constructs the sender's honest f_s, C , and $X=f_s(P)$ in polynomial bit time. Coprime degree preserves the exact N order of X . The sender map/chain is erased and is never a recipient input. Details are in honest-keygen-compatibility.yaml.

Injectivity and actual-message decoding

[DERIVED] If two degree-at-most- D maps to the same exact C agree on exact-order P , then a nonzero difference would have degree both at least N and at most $4D$, impossible. Exact same ciphertext therefore gives the same map and kernel. In an admitted basis,

$$\langle U+sV \rangle = \langle U+tV \rangle$$

forces $s=t \bmod D$: equality gives $U+sV=z(U+tV)$ for a unit z ; compare the U coordinate to get $z=1$, then the V coordinate. The canonical range of $s=m+2^k r$ then gives the same m and r .

[DERIVED] The same coordinate comparison proves $\text{ord}(f(V))=D$; also $f(U)=-[s]f(V)$. Therefore

$$H=[2^{(a-k)}]f(V), G=-[2^{(a-k)}]f(U)=[m]H,$$

with $\text{ord}(H)=2^k$. A checked binary digit loop followed by $[m]H=G$ returns the actual selected message. This theorem remains explicitly conditional on the semantic evaluator contract; local image-order tests alone cannot prove that arbitrary outputs are $f(U), f(V)$. The exact theorem appears in injectivity-decoder-theorem.yaml.

Nearby-object and proves-too-much controls

[DERIVED] Symbolic controls identify one exact separator each:

- $P=0$ or lower-order P : $[N/5]P!=0$.
- $V=0$: $[D/2]V!=0$ and the full-order pairing.
- Exact-order but dependent U, V : $\text{ord}(e_D(U, V))=D$.
- Lower-order $B0$ or H : post-evaluation exact image orders.
- A swapped tuple relative to the frozen key: ordered role/full-byte binding.
- A retagged or isomorphic-only model: canonical full model bytes.
- $x(X)$ only: full signed point encoding.
- A supplied correct evaluator: decoding succeeds, but the evaluator-separation gate forbids calling the supplied object a construction.

The full traces are in `attack-gates.yaml`.

Limitations and obligation map

[PROPOSED / OPEN] Three exact interfaces remain:

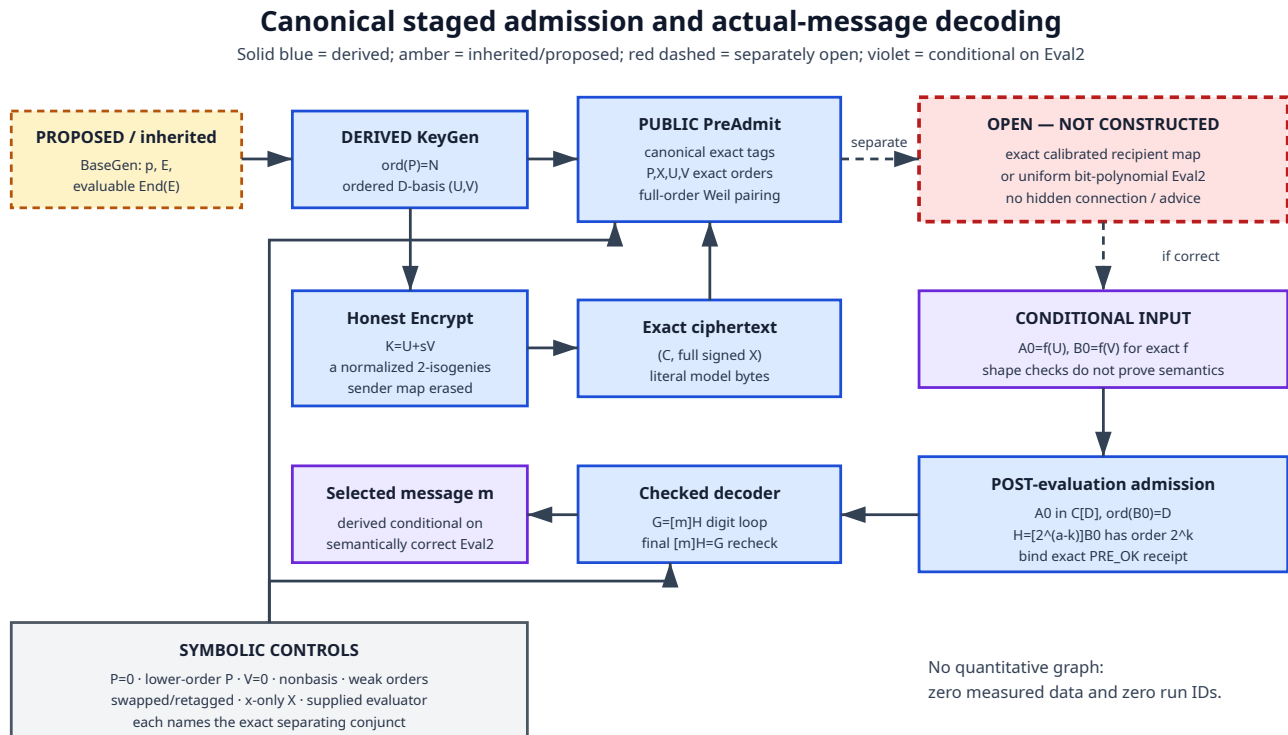
1. autonomous BaseGen for the complete evaluable $\text{End}(E)$ key distribution;
2. exact calibrated recipient-map construction;
3. uniform bit-polynomial Eval2 for $f(U), f(V)$ with all preprocessing, memory, output, success, and failure costs.

All O1--O6 scheme obligations remain open. In particular, a solver followed by this conditional decoder is an attack upper bound, not an IND-CPA reduction. See `scheme-obligation-delta.yaml` and `proof-search-map.yaml`.

No security, novelty, practicality, deployment, finality, family closure, or universal EndRing-PKE possibility/impossibility claim is made.

Visual record

The editable source is `admission-decoder-flow.mmd` and the matching vector rendering is `admission-decoder-flow.svg`. The current PDF is `admission-decoder-report.pdf`.



No quantitative graph applies: there are no measured data, samples, uncertainty intervals, or run IDs. The prior study directory and curated ECDLP frontier were checked conceptually for affected canonical graph content; this interface derivation changes no supported canonical relationship, so no shared graph was edited.

Sources and provenance

- **Internal:** the frozen producer handoff, preparation manifest, and pre-recorded review plan under TASK-20261006-ca9189.

- **Internal:** GOAL-SSI-2fa82a, RQ-SSI-1946c9, and DEC-20261006-3025de for the exact task and claim boundary.
- **Internal:** corrected composition TASK-20261006-97107d, its next-action recommendation, and the prior producer's blind statement, proof map, scheme delta, attack gates, and evaluator reconstruction report.
- **Internal:** the prior visual report research/ssi-endring-pke/2026-10-06-low-degree-evaluator-search/evaluator-search.md, read from the tracked HEAD object because the path is sparse-checkout omitted.
- **Internal methodology:** AGENTS.md, CLAUDE.md, the Idea Generator role, inventor protocol, target profile, scheme-construction contract/template, and the propose-ideas/research-visuals skills.
- **KB, reported tier only:** KN-OPEN-013, KN-TECH-028, KN-TECH-029, KN-TECH-080, KN-LIT-073, KN-LIT-074, and KN-LIT-7642.

External primary retrievals by this worker: none. Novelty status: unverified.